

	LDCW6123 - FUNDAMENTALS OF DIGITAL COMPETENCE FOR PROGRAMMER
	Group Project
	Trimester 2610
	Due date: 3 July 2026 (Friday) 11.59pm

General information:

Group formation	Group members shall not exceed 6 students per group and must be from the same class section. Register the team members in online group registration form.	
Assessment	This group assignment contributes 40% to the total coursework marks.	
Student Learning Time	This assignment shall take 18 hours to complete.	
Submission requirements	Submission	Submission by the group leader only. - Project report (pdf) containing: - - Coversheet - Members Declaration - Table of content (TOC): - Part 1- Innovation Life Cycle Poster - Part 2 – Interactive system (C++) - Recorded video presentation (OneDrive link) - Recorded Video presentation in mpeg/mp4/mov format. Duration Max: 15min Only group leaders will make the submission. You will follow your respective tutor submission procedure. Name your report as; <i>Ldcw6123_project_section_group leader name(student id).pdf</i>
	Cover page	Please use the attached coversheet.
	Group member's declaration form	Please use the attached form.
	Assessment rubrics	Included for reference. You do not need to include the rubrics into the report.

Mapping of assignment learning outcomes to subject learning outcomes:

Project Learning Outcomes Upon completion of this project, students should be able to:		Course Learning Outcomes Upon completion of this subject, students should be able to:
1.	Develop a poster and program showcasing the technology from an innovation lifecycle.	CLO3: Build a basic interactive program with digital content.

Project Title: Innovation Technology Life Cycle Poster and Interactive Program

Group member: Min 4 – Max 6 student per group.

Instructions:

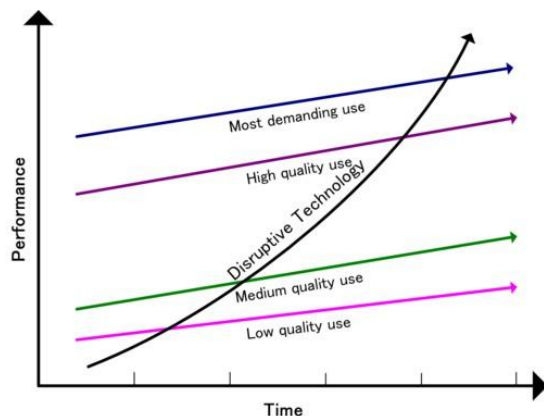
students will work together to design and develop part 1 and 2 of the project. Each group member will collaborate online to complete the projects.

PART 1 (INNOVATION TECHNOLOGY LIFE CYCLE POSTER) – 20%

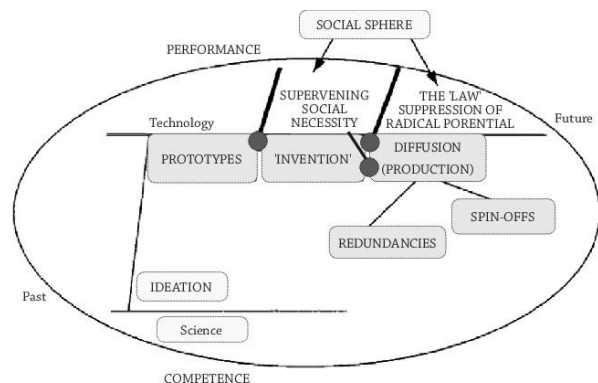
Choose a **specific scientific or technological innovation** and trace the development life cycle on a poster. Your chosen technology may be something local (like a Aerodyne Drones - Malaysian global leader in drone-based solutions for agriculture, inspection, and security) or international (e.g.: Netflix, TikTok Social Commerce). This poster should be presented as either an A3 poster made into a PDF, or as PowerPoint slides. The “innovation model” you choose will depend on the innovation you want to trace. If the innovation was “disruptive” you might choose **Clayton Christensen’s model**. If the innovation *met a pre-existing need in society* (e.g., poster for the telegraph) you might choose **Brian Winston’s model**. There is a template for both models provided along with this project assignment on eBwise.

Please check with your lecturer if you are unsure about the innovation/invention you have chosen.

Depending on the innovation you choose, and whether or not that product met a pre-existing social need or alternatively created/disrupted a market, you may choose one of the following models to apply. The steps to identify are different in each case: if you choose **Christensen’s model** you will also need to show which market was ‘disrupted’. If you choose **Winston’s model** you will need to identify the social needs (“supervening social necessities”) that aided invention and adoption.



Clayton Christenson’s Disruptive cycle model
- Innovation technology



Brian Winston’s cycle model –
Innovation technology

Once you have identified the different steps in the development of your innovation, you can create the flow chart in Canva/Adobe software or in PowerPoint, or in any software program you have at hand. You will not be marked on your “design” abilities, **but the flow of the innovation should be clear**. You will be assessed on your ability to identify the correct steps in the innovation life cycle, your understanding of the

history of your specific innovation, and the historical accuracy of those steps. Make sure you choose an innovation where you can determine the different steps. Include this poster in your PDF report submission.

Criteria:

- (1) A clear schematic representation of the innovation life cycle
- (2) Evidence of research – make your steps historically accurate
- (3) Evidence that you understand the “innovation life cycle” you have chosen
- (4) Evidence that you understand its social impact or the social necessity
- (5) Place referencing (APA Style)

PART 2 (INTERACTIVE PROGRAM) – 20%

After completing Part 1, where you researched and discussed your chosen technology using either the Disruptive Innovation or Winston’s cycle model, you will now proceed to Part 2 by designing a simple interactive program using any prefer computer language (e.g.: C++ or any).

General Steps

1. Connect with Your Part 1 Topic
 - Your chosen technology in Part 1 (e.g., Netflix, Uber, e-commerce platform) should be used as inspiration for your interactive program.
2. Decide the purpose of your program
 - Think of a real-world feature or function relevant to your technology. Examples:
 - **Netflix:** Movie finder or recommendation based on user preference.
 - **Ride-hailing:** Simple fare calculator.
 - **E-commerce:** Product suggestion or shopping cart summary.
3. Outline Inputs and Outputs
 - Plan what the user will enter (genre, distance, product type) and what the program will display (suggestion, calculation, confirmation).
4. Design Logic
 - Use if/else or switch statements to process user choices and produce results.
5. Document Development Progress with Git Log
 - As part of your submission, include a trace of your project’s development history using Git.
 - Make regular commits at each stage (e.g., after outlining inputs/outputs, designing logic, coding features).
 - At the end, run ``git log --online --graph`` and include a screenshot or text output in your submission.

Example:

If you chose Netflix and the Disruptive Innovation model:

Title program: Movie Recommendation Assistant
Inputs: Movie genre preference
Outputs: A suggested movie title and a brief description

- (1) Include a screenshot of the codes and GitLog
- (2) Paste the link to the online compiler (code editor) and GitLog
- (3) Include part 2 in the PDF report
- (4) Record a group presentation video. Paste the link to your video in the PDF report.

Assessment Rubric

Part 1: Innovation Technology Life Cycle Poster (40 marks)

Criteria/ Level	Excellent	Good	Satisfactory	Weak	Poor
Research & Background	Comprehensive explanation of technology, highly accurate historical timeline, multiple credible sources used.	Clear explanation, mostly accurate history, sufficient supporting sources.	Basic explanation, some inaccuracies, limited research evidence.	Incomplete explanation, unclear or incorrect history.	Very minimal or incorrect content, no evidence of research.
10 marks	9-10	7-8	5-6	3-4	2-0
Model Explanation & Application	Model clearly explained, all stages accurately applied to the chosen technology with strong justification.	Model explained well, most stages correctly applied.	Basic explanation, partial application of model.	Weak understanding, incorrect or unclear application.	Model not explained or wrongly applied.
12 marks	10-12	7-9	5-6	3-4	0-2
Mapping & Impact Analysis	Logical and well-structured flow, strong analysis of impact/disruption with relevant examples.	Clear flow, good discussion of impact.	Basic flow, limited impact discussion.	Poor flow, unclear or weak analysis.	No clear mapping, no impact discussion.
10 marks	9-10	7-8	5-6	3-4	2-0
Critical Thinking	Insightful analysis, original ideas, strong understanding.	Some analysis, shows understanding.	Limited insight, mostly descriptive.	Minimal thinking, superficial discussion.	No critical thinking shown.
4 marks	4	3	2	1	0
Clarity & Organization	Very clear, well-structured, easy to follow.	Generally clear with minor issues.	Some unclear sections.	Poor organization.	Very confusing presentation.
4 marks	4	3	2	1	0

Part 2: Interactive C++ Program (40 marks)

Criteria/ Level	Excellent	Good	Satisfactory	Weak	Poor
Program Logic & Functionality	Fully functional, correct logic, handles multiple scenarios accurately.	Mostly functional, minor logic issues.	Basic functionality, limited logic.	Many errors, partially working.	Not functioning.
10 marks	9-10	7-8	5-6	3-4	2-0
Relevance & Integration	Strong integration with Part 1, meaningful and realistic application.	Good relation to topic.	Some relevance.	Weak connection.	No relation to Part 1.
10 marks	9-10	7-8	5-6	3-4	2-0
Code Quality & Documentation	Clean, well-structured code, meaningful variable names, proper comments.	Mostly clean, some comments.	Basic readability, minimal comments.	Poor structure, unclear naming.	No structure or unreadable code.
5 marks	5	4	3	1-2	0
User Experience & Testing	Clear instructions, handles errors, tested with multiple cases.	Good usability, minor issues.	Basic usability, limited testing.	Confusing interface, minimal testing.	No testing or usability consideration.
5 marks	5	4	3	1-2	0
Git log & Development Process a) Commit Quality	Frequent commits, clear, meaningful messages.	Regular commits, mostly meaningful.	Some commits, basic messages.	Few commits, unclear messages.	No Git usage.
5 marks	5	4	3	1-2	0
Git log & Development Process b) Git Log Presentation	Clear, well-presented log showing development progress.	Mostly clear log.	Basic log shown.	Incomplete log.	No log included.
5 marks	5	4	3	1-2	0

Video recording presentation and requirements/compliance (20 marks)

Criteria/ Level	Excellent	Good	Satisfactory	Weak	Poor
Content Coverage & Explanation	Clear, complete explanation, strong understanding; effective demo.	Good explanation, minor gaps.	Basic explanation.	Incomplete explanation.	Very unclear or missing content.
10 marks	9-10	7-8	5-6	3-4	2-0
Compliance & Documentation	Follows all submission instructions, includes all required documentation.	Minor missing elements.	Some missing components.	Many missing requirements.	Not compliant.
5 marks	5	4	3	1-2	0
Originality & Integrity	Highly original, properly cited, no issues.	Mostly original, minor citation issues.	Limited originality.	Significant similarity issues.	Plagiarism detected.
5 marks	5	4	3	1-2	0

Total Marks Calculation:

The total marks for this assignment will be calculated by adding up the marks awarded for both part and video/requirements based on the provided evaluation criteria. The final grade will reflect the cumulative score out of 100, based on the following breakdown:

To calculate the total marks for the assignment:

- Part 1: Score each part out of a total of **40 marks**.
- Part 2: Score each part out of a total of **40 marks**.
- Video recording presentation & requirements: Score each part out of a total of **20 marks**.

Total Marks Possible: **100 marks**

Reminders:

Plagiarism

- Plagiarism is a serious offence.
- Plagiarism includes the following behaviours (but not limited to):
 - copy another student's work.
 - fail to properly cite other people's work or give proper credit to the original source.
 - using photos, illustrations or online materials downloaded from websites without permission or consent of the original owner of the materials.
 - hire or ask another individual to complete the assignment for you.
 - copy too many words or ideas from one or two sources, that makes up a significant portion of your work, even with proper citations.
- The lecturer has the right not to accept submission of plagiarized or duplicated work.

Group work

- Each group member must actively participate in the completion of the assignment. Free riding is unethical and extremely unfair to fellow group members. **Free riders will be awarded zero mark.**
- Should any member be found not contributing to the assignment, the lecturer has the right to change the specific group's marks to individual marks.
- Each group member is expected to keep the contact numbers and e-mail addresses of the other members of the group.
- The group leader has the responsibility to ensure that all group members play a part in the completion of the assignment. The group leader is also responsible for ensuring that all members' names are written on the final submission.
- It is the responsibility of each group member to ensure that the final submission is complete and of an acceptable standard. Should any errors or omissions occur in the final submission, each group member is held accountable for negligence – in other words, failure to perform their responsibilities as expected.

Late submissions

- Late submissions will incur a penalty of 10% per day, up to a maximum of five days, after which the assignment will no longer be accepted.
- Additionally, if a second chance is granted, a further penalty of 10% will apply for one additional opportunity.
- Exceptions may be made for late submissions due to serious and valid circumstances (e.g., illness or family emergency), provided appropriate supporting evidence is submitted.



ASSESSMENT COVERSHEET

Trimester 2610

Attach this coversheet as the cover of your submission. All sections must be completed. **Section**

A: Submission Details

Programme : _____
Course Code & Name : _____
Course Lecturer(s) : _____
Submission Title : _____
Deadline : Day _____ Month _____ Year _____ Time _____

Penalties : • Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly.

- Late submissions will incur a penalty of 10% per day, up to a maximum of five days, after which the assignment will no longer be accepted.

Section B: Academic Integrity

Tick (✓) each box below if you agree:

- | | |
|--------------------------|--|
| ☐ | I have read and understood the policy on Plagiarism in University Rules & Regulations. |
| ☐ | This submission is my own, unless indicated with proper referencing. |
| ☐ | This submission has not been previously submitted or published. This submission follows the requirements stated in the course. |
| ☐ | |

Section C: Submission Details

Student ID(s)	Student Name(s)	Designation
		(Group leader)

Declaration by group leader

I hereby declare that all group members' names are correctly included in the above section. I hold a copy of this assignment which I can produce if the original is lost or damaged. I certify that no part of this assignment has been copied from any other student's work or from any other source except where due acknowledgement is made in the assignment.

Group leader's signature: _____

Group leader's name : _____

Group leader's student ID: _____

Date : _____

Group member declaration

*(Each group member, including the **group leader**, must individually fill up and submit this form)*

For the purpose of completing this assignment, I have performed the following tasks (please list):

I hereby declare that I have assessed this submission, and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature: _____

Group member's name : _____

Group member's ID : _____

Date : _____

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